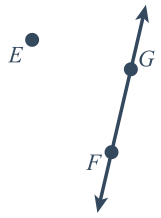
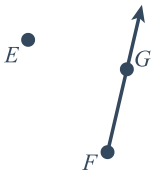
Review: Points, lines, and planes

1 Name the line, line segment, or ray in each diagram below using geometric notation.

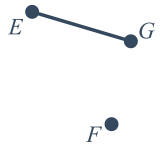
a



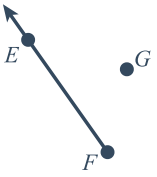
b



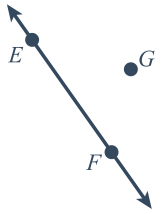
c



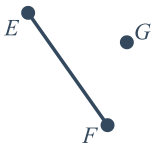
d



e

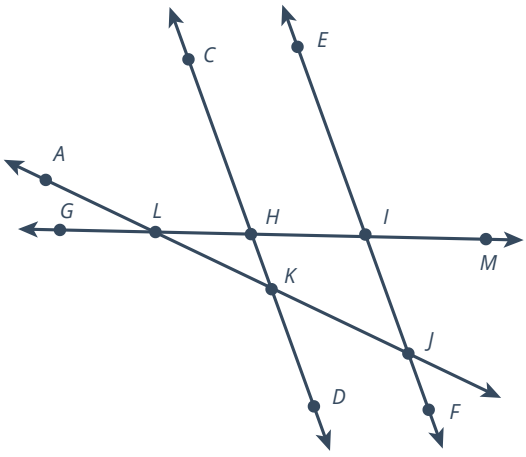


f



2 Name three different pairs of collinear points in each diagram below.

a



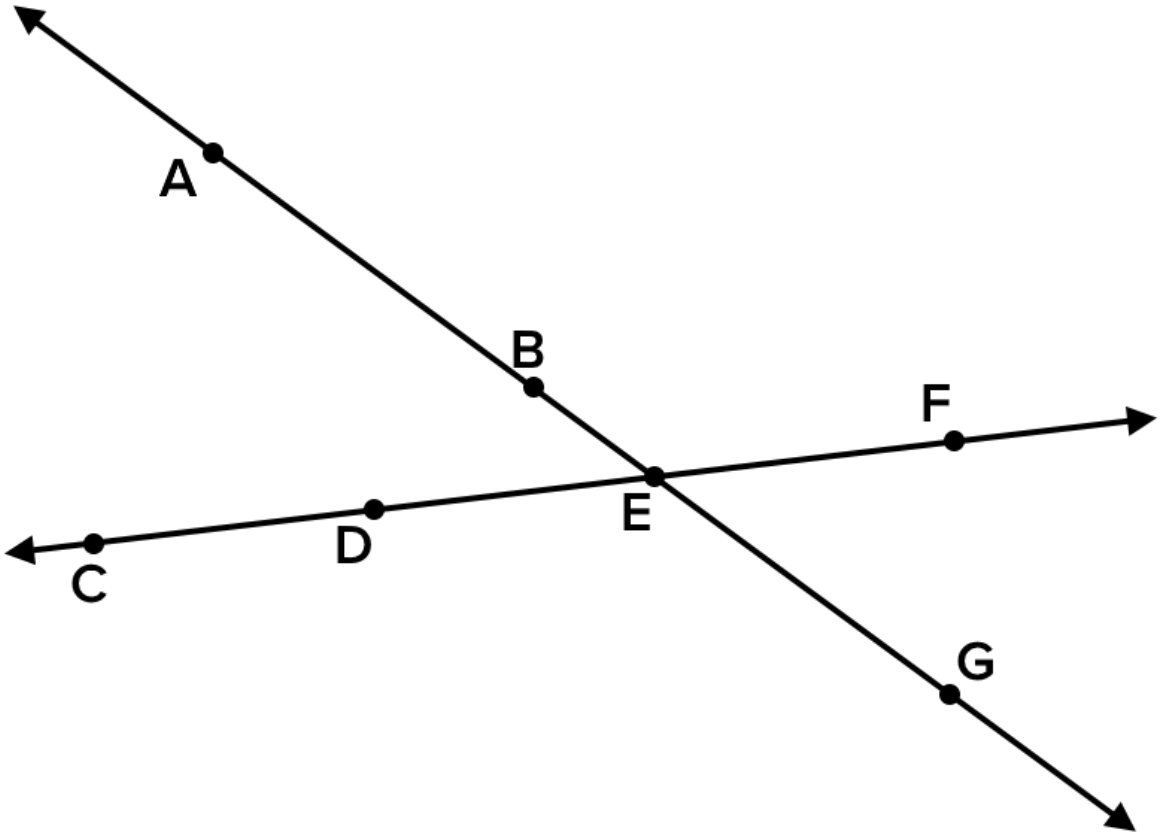
b



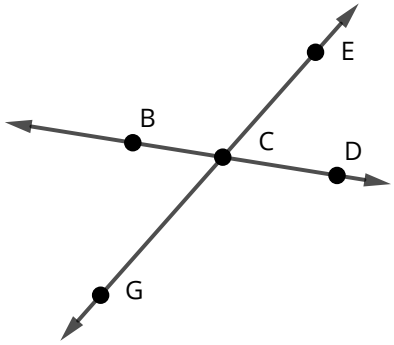
3 What is another way to define the line $\overleftrightarrow{CD}$?

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4 What is another way to describe the same set of points as $\overline{FD}$?



5 Determine whether the statement is true or false based on the diagram.



a Point C is on $\overleftrightarrow{GE}$.

b Point C is on $\overleftrightarrow{BD}$.

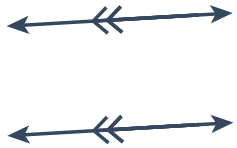
c The intersection of $\overleftrightarrow{GE}$ and $\overleftrightarrow{BD}$ is point C.

d $\overleftrightarrow{GE}$ and $\overleftrightarrow{BD}$ must be perpendicular because they intersect.

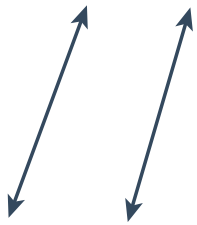
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6 In each diagram below, state whether the lines are parallel or not parallel.

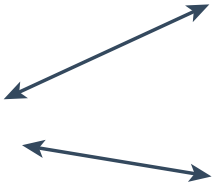
a



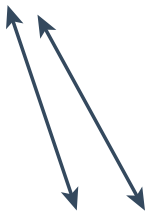
b



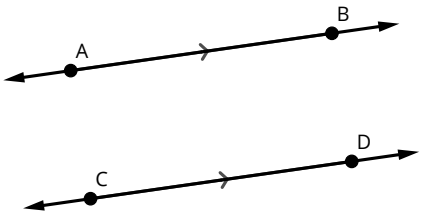
c



d



7 Consider the diagram below.



State whether each of the following statements is true, false, or if there is not enough information to decide.

a $\overline{AB} \parallel \overline{CD}$

b $\overline{AB} \cong \overline{CD}$

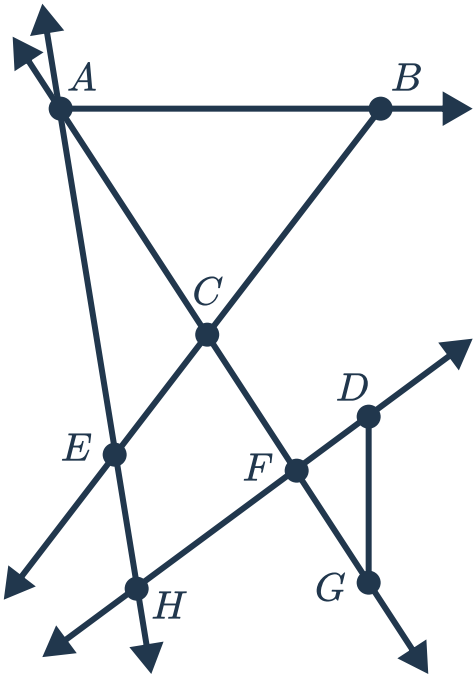
c $\overleftrightarrow{AB} \parallel \overleftrightarrow{CD}$

d $\overrightarrow{AB} \parallel \overrightarrow{CD}$

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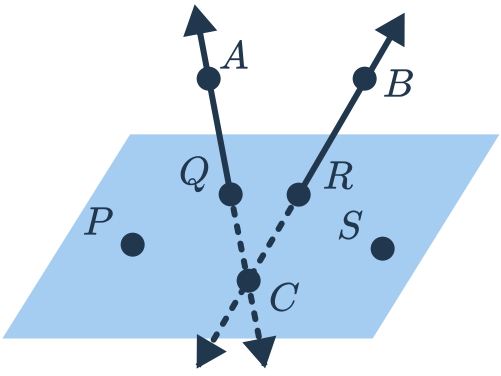
Additional questions

8 Using proper notation, identify the following features of the diagram shown:



- **a** A set of three collinear points
- **b** A line passing through point F
- **c** A ray starting from point A
- **d** Another name for the ray $\overrightarrow{BE}$

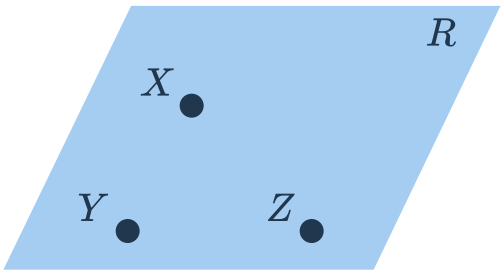
9 Using proper notation, identify the following features of the diagram shown:



- **a** A line that is contained in plane PQR
- **b** A line that intersects, but is not contained in, plane PQR
- **c** Four coplanar points
- **d** Three collinear points
- **e** An angle that is contained in the plane
- **f** An angle that is not contained in the plane
- **g** A pair of opposite rays

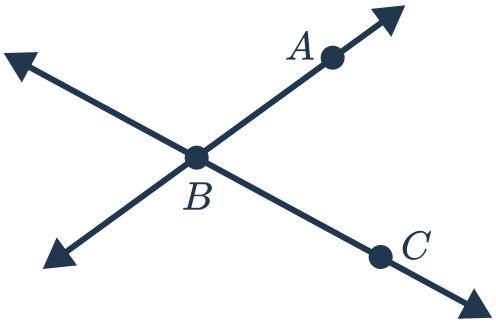
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10 Name the geometric figure shown below in two different ways:

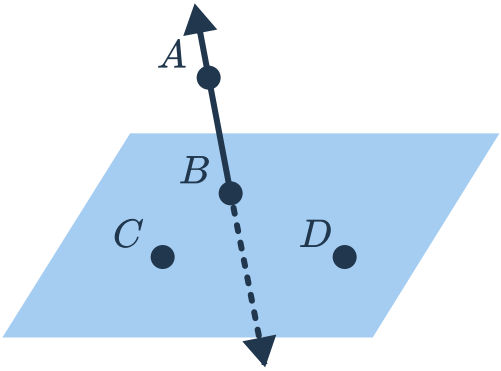


11 State the intersection of the two geometric figures shown in each of the following diagrams:

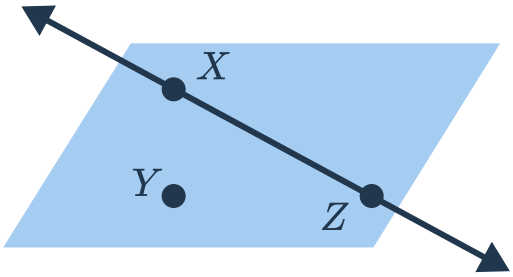
a Lines $\overleftrightarrow{AB}$ and $\overleftrightarrow{BC}$



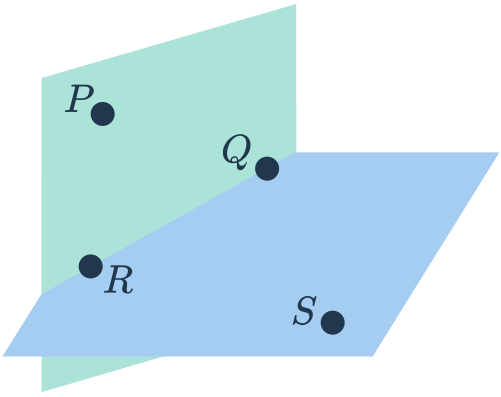
b Line $\overleftrightarrow{AB}$ and Plane BCD



c Line $\overleftrightarrow{XZ}$ and Plane XYZ



d Planes PQR and QRS



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12 Answer the following questions:

- **a** Draw a ray that has point A as an endpoint.
- **b** Write a name for the ray you have drawn.
- **c** How many points did you need to use to name the ray?
- **d** When naming the ray, does the order of the points matter? Explain your answer.

13 Lines $\overleftrightarrow{PQ}$ and $\overleftrightarrow{RS}$ meet at point M.

- **a** Draw a diagram that shows this.
- **b** Identify a pair of opposite rays in the diagram.

14 Planes DEF and EFG have common points of E and F.

- **a** Draw a diagram that shows this.
- **b** Name the full intersection of the two planes.

15 Draw a diagram that shows $\overline{AB}$ contained in plane P, with $\overleftrightarrow{CB}$ intersecting (but not conatined in) the plane.

16 Determine whether each of the following statements are true or false. If true, draw an example. If false, draw a counter example.

- **a** Through any two points, there is exactly one line.
- **b** Two distinct pairs of points always form two different lines.
- **c** If two points are both contained in a plane, then the line that they form is also entirely contained in that plane.
- **d** For any three collinear points, there is exactly one plane that contains them.
- **e** If three points are collinear, they cannot be contained in a single plane.
- **f** For any two distinct lines, there is a plane which contains them both.

17 For each of the following geometric figures, determine a real life example which is a model for the figure. Are there any limitations or restrictions on the real life example?

a A ray **b** A line **c** A plane **d** Skew lines

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